



Low-crystallinity iron oxide minerals generated from iron powder applied to rice paddy enhanced microbial nitrogen fixation in soil

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Abstract

Biological nitrogen (N) fixation is crucial for sustaining the N fertility in rice paddy soils. The N-fixing activity of paddy soils is higher than that of upland soils and increases during the flooded period as soil reduction progresses. Previously, we identified that N fixation in paddy soils is actively driven by iron (Fe)-reducing bacteria belonging to *Deltaproteobacteria*, which use Fe oxides as electron acceptors for respiration. Applying oxidized Fe powder to paddy soil enhanced the energy metabolism of these microorganisms, thereby increasing their N fixation activity. However, the Fe-oxide minerals responsible for this enhancement remain unknown. In this study, we identified the Fe-oxide minerals generated from Fe powder applied to paddy soils and evaluated their effects on soil N fixation. Structural analyses using XRD, SEM (-EDS)/TEM, and μ XRF-XAFS showed that ferrihydrite and goethite predominated near the Fe-oxide particles, whereas lepidocrocite was more abundant farther from the particles. Some ferrihydrite incorporated Si, which was likely obtained from the surrounding soil. Of the identified Fe-oxide minerals, ferrihydrite and lepidocrocite significantly enhanced the N-fixing activity in paddy soil microcosms, suggesting that these low-crystalline Fe oxides were responsible for the enhanced N fixation observed in previous field studies. This study shows that the mineralogical composition of Fe oxide is essential for regulating microbial N fixation, and that Fe oxidation conditions should be optimized to selectively generate ferrihydrite and lepidocrocite, offering a promising strategy for increasing N fertility in paddy soils.

Keywords Iron application · Iron oxides · Iron-reducing bacteria · Nitrogen fixation · Rice paddy soil

Introduction

Rice (*Oryza sativa* L.) is a staple crop in Asian countries. Chemical nitrogen (N) fertilizers have been widely applied to rice paddies to boost yield and meet rising regional rice demand. However, the long-term overuse of N fertilizers has caused serious environmental N loading, leading to nitrate leaching, ammonia volatilization, and nitrous oxide emissions from paddy soils (Shen et al. 2022). These processes

contribute to air pollution, global warming, the eutrophication of aquatic systems, and nitrate contamination of groundwater (Ju et al. 2009; Liu et al. 2013; Zhang et al. 2013, 2015). Thus, reducing the use of N fertilizers is essential to support environmentally sustainable rice production.

Biological N fixation is a natural alternative to chemical fertilizers for supplying N to paddy fields (De Bruijn 2015). The N-fixing activity of paddy soils is higher than that of upland soils (Yoshida and Ancajas 1973), and is elevated during the flooded period when rice grows (Masuda et al. 2023). Therefore, relatively stable rice yields can be maintained in the absence of synthetic nitrogen inputs (Ladha et al. 2016). This nitrogen-fixing activity is critical for maintaining soil nitrogen fertility and crop productivity and has conventionally been dominated by cyanobacteria and rhizobia (Mårtensson et al. 2009, Shapleigh 2006). However, these conclusions were largely based on cultivation-dependent approaches or PCR-based analyses that target specific *nif* genes, which may underestimate uncultured or primer-mismatched microorganisms.

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Previously, we conducted shotgun metatranscriptomic sequencing of soil RNA to identify key N-fixing microorganisms in paddy soils without PCR amplification bias. We found that transcripts of the nitrogenase gene (*nifD*) were predominantly derived from members of the families *Anaeromyxobacteraceae* and *Geobacteraceae* (Masuda et al. 2017). N-fixing activity has also been confirmed in strains isolated from paddy soils belonging to these groups (Masuda et al. 2020; Xu et al. 2019). Furthermore, global metagenomic analyses have demonstrated that *nifD* genes derived from these bacteria are enriched in paddy soils and sedimentary environments worldwide (Masuda et al. 2024b).

These iron (Fe)-reducing bacteria use Fe oxides as electron acceptors during anaerobic respiration. Therefore, we conducted experiments to test whether the addition of Fe oxides to paddy soils stimulates the respiratory activity of these bacteria and enhances their N-fixing activity. Pure (zero-valent) Fe powder was applied to the soil surface and allowed to oxidize for 1 week under field conditions to increase the amount of Fe oxides in the soil and form trivalent Fe oxides, which were then incorporated into the soil via plowing. The N-fixing activity of the soil was markedly higher in Fe-oxide-treated plots than in the control plots during rice cultivation (Masuda et al. 2021). Furthermore, ^{15}N -isotope ratio mass spectrometry analysis and ^{15}N -DNA stable isotope probing revealed that this enhancement was driven by the activity of the Fe-reducing bacteria of the families *Anaeromyxobacteraceae* and *Geobacteraceae* (Zhang et al. 2023). These findings suggest that adding Fe oxides to soil increases microbial N fixation and may improve N fertility.

However, the specific forms of Fe oxide generated from Fe powder remain unclear because we previously applied and oxidized pure Fe powder under natural field conditions. Fe in paddy fields is reduced to dissolved forms during repeated flooding and oxidized to hydroxide minerals during the drainage, and this process is repeated (Wang et al. 2022). These redox cycles form Fe hydroxide minerals with diverse and complex morphologies in paddy soils. For example, the initial mineral form of Fe hydroxides is generally low-crystallinity or amorphous ferrihydrite. As the oxidation proceeds, ferrihydrite transforms into goethite, which is a more crystalline Fe hydroxide mineral. Alternatively, in environments rich in impurities such as silicon (Si), the transformation of ferrihydrite into goethite is inhibited, and it is more likely to transform into lepidocrocite (Schulz et al. 2022).

Ferrihydrite and lepidocrocite are low-crystallinity Fe minerals with large specific surface areas (Poulton and Canfield 2005). Generally, the larger the specific surface area of an Fe mineral, the stronger the potential for microorganisms

to exchange electrons with these minerals through direct contact with their cell bodies, nanowires, or bioshuttles (Shi et al. 2009; Reguera et al. 2006; Roden et al. 2010). Goethite is a highly crystalline Fe mineral with a small specific surface area and low microbial accessibility (Bray et al. 2017). As described above, Fe minerals with different crystal structures and specific surface areas are expected to have different effects on the Fe utilization and N-fixing activities of Fe-reducing bacteria; however, the extent of these effects remains unclear.

Therefore, we aimed to elucidate the Fe-oxide crystal structures generated from pure Fe powder in paddy soils and determine their effects on N fixing. Identifying the Fe-oxide crystal structures that most effectively enhance N fixation will help generalize and optimize this Fe method for application in agricultural systems.

Materials and methods

Sampling and characterizing soil

Soil was collected from the plow layer of a paddy field at the Niigata Agricultural Research Institute, Nagaoka, Niigata prefecture, Japan (37°44' N, 138°87' E) in April 2021. The physicochemical characteristics of the soil are described in our previous report (Masuda et al. 2021). The soil was sieved through a 2 mm mesh and stored at 4 °C until use. The rice straw residue in the soil, which had been incorporated into the field soil, was removed via sieving. After sieving, the soil moisture content was approximately 37%. This soil was collected from the same field as that used in Zhang et al. (2023), where nitrogen fixation by iron-reducing bacteria was reported to be active.

Oxidation of pure Fe powder in the soil microcosm

A 20 g paddy soil sample was placed in a 150 mL plastic beaker, and 0.1 g of pure Fe powder for agricultural use (iron powder for agriculture, JFE Steel Corporation, Tokyo, Japan) was applied to the soil surface. The prepared soil microcosm was incubated at 30 °C for 7 days to ensure the oxidation of the Fe powder. The soil surface was moistened every 2–3 days with distilled water (Elix Essential UV3, Merck KGaA, Darmstadt, Germany) to prevent drying. Post-incubation, Fe-oxide powder and adjacent soil were collected using a stainless-steel spatula and air-dried.

X-ray diffraction (XRD)

An Fe-oxide powder sample was obtained from the soil surface and homogenized using a mortar and pestle for

subsequent structural analysis using XRD. The sample was carefully separated from the soil, retaining only a minimal amount of adhering soil because the amount of Fe oxide was small compared with that of the clay minerals and quartz typically found in natural soils. The isolated powder was analyzed using powder XRD.

We employed Cu K_{α} radiation (40 kV, 30 mA, with a Ni filter to attenuate Cu K_{β}), and a Si strip-type one-dimensional detector (D/teX Ultra, Rigaku, Tokyo, Japan) mounted on a RINT-Ultima⁺ goniometer (Rigaku). The divergence slit was set at a divergence angle of $1/4^{\circ}$. The goniometer was operated at a scan speed of 1° (2θ)/min with a sampling interval of 0.02° (2θ), covering a 2θ range of 10 – 50° .

Scanning electron microscopy (SEM)/SEM-Energy-dispersive X-ray spectroscopy (EDS)

The Fe-oxide powder was coated with a Pt–Pd alloy to prevent sample charging during morphological observations using SEM and SEM-EDS. The Fe-oxide powder was examined using cold-type field-emission SEM (S-4500, Hitachi High-Tech Corporation, Tokyo, Japan) operated at 2 kV. The Fe-oxide powder was quantitatively analyzed using EDS (Noran System7, Thermo Fisher Scientific Inc., MA, USA) combined with SEM. The EDS spectra were acquired at an acceleration voltage of 15 kV.

Transmission electron microscopy (TEM)/scanning TEM (STEM)

The SEM samples were thinned using conventional grinding to achieve electron transparency using a focused ion beam system (FB-2100, Hitachi High-Tech Corporation) for TEM and STEM. Bright-field images and electron diffraction patterns were recorded using TEM (JEM2800F, JEOL, Ltd. Tokyo, Japan) and a CCD camera (Gatan Orius, Ametek Co., Ltd., PA, USA). The local chemical compositions were analyzed using an EDS detector (JED-2300) combined with TEM.

μ XAFS-XRF

The Fe-oxide powder, along with the soil particles, was embedded in a nontoxic, room-temperature-curing epoxy resin (EpoxiCure J, Buehler, IL, USA) for elemental mapping and X-ray absorption near-edge structure (μ XAFS-XANES) analysis using fine-structure micro-X-ray absorption. The samples were sectioned with water after being embedded in an Isomet automatic precision cutting machine (Buehler, IL, USA). A thin-section sample was mounted on a glass slide with beeswax and dry polished. Fe K-edge μ XRF-XANES analyses were conducted at

the beamline BL-4 A of KEK Photon Factory in Tsukuba, Japan. A thin sample was fixed on a sample holder oriented 45° to the beam. The beam was monochromatized with a Si (111) double-crystal monochromator and focused to 5 (vertical) $\times$ 5 (horizontal) μm with a Kirkpatrick–Baez mirror system. The Fe μ XANES spectra of the reference materials and the soil sample were measured in transmission mode. The energy of Fe was calibrated using the pre-edge peak maximum of hematite, which was fixed at 7.113 keV.

A thin-section sample was fixed to an XY stage and controlled using X-Y pulse motors for μ XANES measurements. The μ XRF maps of the soil grains for Fe were collected to determine the points of interest in the sample via moving the sample with the pulse motor, and then the XANES spectrum was measured at the point of interest. The μ XANES spectra were processed using REX2000, a data analysis software package (Rigaku). Standard Fe mineral samples (magnetite, goethite, lepidocrocite, ferrihydrite, and Si-containing ferrihydrite) were used in this study. Standard samples of divalent and trivalent Fe in 2:1 phyllosilicate, biotite, and montmorillonite were used. Si-containing ferrihydrite and lepidocrocite were synthesized in our laboratory, as described below.

Preparing Fe minerals

Six Fe-bearing minerals were used in this study. Si-containing ferrihydrite, ferrihydrite, lepidocrocite, and goethite were used for the Fe addition experiments. Magnetite, biotite, and montmorillonite were used for XANES fitting along with these four Fe minerals.

Si-containing ferrihydrite, ferrihydrite, and lepidocrocite were synthesized as follows: All reagents were prepared with double deionized water ($>18.2\text{ M}\Omega\cdot\text{cm}$, Synergy UV, Merck KGaA).

Ferrihydrite was synthesized using the method described by Schwertmann and Cornell (1991). A 0.2 M ferric nitrate ($\text{Fe}(\text{NO}_3)_3\cdot 9\text{H}_2\text{O}$) solution was titrated with 1 M NaOH to $\text{pH } 7.5 \pm 0.3$ under vigorous stirring. Si-containing ferrihydrite was synthesized as described by Schulz et al. (2022). The synthesis procedure was the same as that for ferrihydrite; however, Si-containing NaOH was added to the ferric nitrate solution to achieve a Si/Fe ratio of 0.33. Si-containing NaOH was prepared by adding sodium metasilicate granules (Na_2SiO_3) to a 1 M NaOH solution. The pH of the suspension was adjusted to $\text{pH } 7.5 \pm 0.3$ with silicate-free NaOH. The Si-containing and pure ferrihydrite precipitates were centrifuge-washed with double deionized water until the salt concentration was below $50\text{ }\mu\text{S/cm}$, freeze-dried, and homogenized with a mortar and pestle.

Lepidocrocite was synthesized as described by Schwertmann and Cornell (1991). A 0.2 M Fe(II) solution (300 mL)

prepared from FeCl_2 was titrated to pH 6.7–6.9 with 1 M NaOH (approximately 120 mL) and oxidized under vigorous stirring and gentle purging with air (approximately 400 mL/min). The precipitate was washed overnight on a membrane filter, freeze-dried, and homogenized. The crystal structures of all the minerals were confirmed using XRD. The scattering slit was set to a scattering angle of $1/4^\circ$, scan speed of $10^\circ (2\theta)/\text{min}$, and sampling interval of $0.01^\circ (2\theta)$, covering a 2θ range of $10\text{--}50^\circ$ for the XRD measurement.

We purchased magnetite (Kojundo Chemical Laboratory, Co., Saitama, Japan), goethite (Strem Chemicals, MA, USA), biotite (Nichika Inc., Kyoto, Japan), and montmorillonite (SWy-3, Source Clays Repository, WY, USA).

Measurement of N fixation in soil microcosms amended with synthesized Fe-oxide minerals

The soil microcosms simulating paddy soil conditions were prepared as follows: Rice straw segments (~5 mm) were added to paddy soil (50% moisture content) to construct the soil microcosms. Rice straw and Fe were added at 1% and 0.5% of the soil (w/w), respectively. Si-containing ferrihydrite, ferrihydrite, lepidocrocite, and goethite were added to the soil as Fe minerals following previously reported methods (Masuda et al. 2021). We placed 10 g of each of the Fe-mixed and control soil samples in a serum bottle and sealed it with butyl rubber and a plastic cap. The soil microcosms

were incubated at 30°C , and soil samples were obtained on days 6 and 9 of incubation, as previously described (Masuda et al. 2021). Because the microcosms were maintained as closed systems until measurement, neither oxygen nor water were lost from the system. Each treatment had three replications.

The N-fixing activity of the soil samples from the microcosm was measured using a standard acetylene reduction activity assay based on C_2H_2 reduction to C_2H_4 by nitrogenase (Postgate 1972). Approximately 3 g of soil was collected from each microcosm in triplicate and the average of the three samples was used as the acetylene reduction activity of the microcosm. Each soil sample was placed in a serum bottle capped with a butyl rubber stopper and sealed with an aluminum cover. The gaseous phase was replaced with $\text{Ar}/\text{C}_2\text{H}_2$ gas (90:10, v/v), and the system was incubated at 30°C for 24 h. Gas samples were collected after incubation from the gaseous phase in bottles, and the C_2H_4 concentration was measured using a gas chromatograph (GC-17 A, Shimadzu, Kyoto, Japan) equipped with a fused-silica column (Rt-U PLOT, Restek, PA, USA) (Nakajima et al. 2012).

Results and discussion

Mineral composition of Fe-oxide powder

The crystal structure of the sample obtained from the oxidation of the pure Fe powder was analyzed using powder XRD. Diffraction peaks corresponding to goethite and lepidocrocite were detected in the Fe-oxide powder sample (Fig. 1a). In contrast, the original pure Fe powder sample exhibited strong diffraction peaks for metallic Fe and weak peaks for magnetite (Fig. 1b). These peaks were absent in the oxidized Fe powder sample. The magnetite in the pre-oxidized Fe powder sample could have been produced via the partial oxidation of the Fe powder due to exposure to atmospheric oxygen during storage. The disappearance of the metallic Fe and magnetite peaks, along with the appearance of goethite and lepidocrocite peaks in the oxidized Fe powder sample, indicate that pure Fe powder and magnetite were morphologically transformed into these Fe-oxide minerals during oxidation in the soil microcosm. Residual pure Fe powder may have persisted after oxidation and was encapsulated by newly formed Fe oxides, which would have been undetectable during analysis.

Ferrihydrite is an amorphous, low-crystallinity Fe-oxide mineral commonly found in paddy soils (Grigg et al. 2022). However, the presence of ferrihydrite in the Fe-oxide powder was unknown because its low crystallinity prevented its detection using XRD (Furukawa et al. 2002). In response,

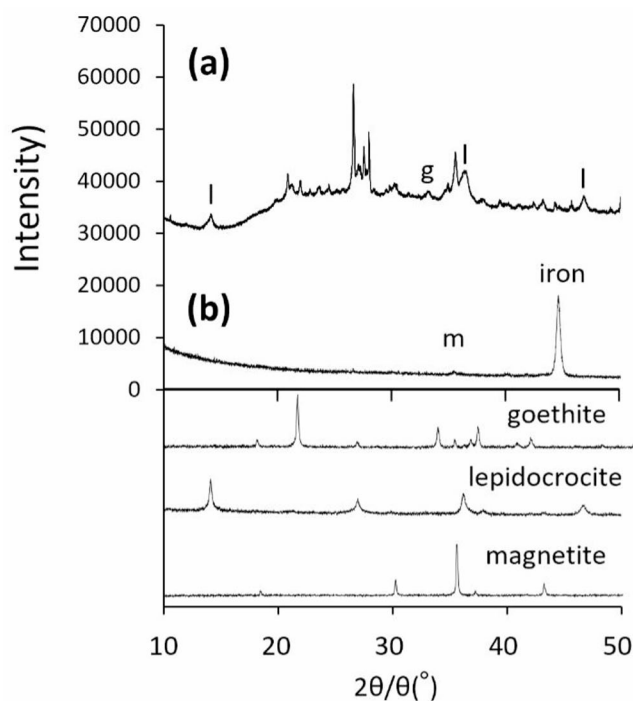


Fig. 1 XRD patterns of Fe-oxide (a) and pure Fe (b) powder with standards (goethite, lepidocrocite, and magnetite). Each peak in (a) and (b) is labeled with the peak name of the Fe mineral: g = goethite, l = lepidocrocite, m = magnetite, and Fe = metallic Fe

we separated Fe-oxide powder from the soil (Fig. 2a) and examined the powder for the presence of low-crystallinity or amorphous mineral structures. SEM revealed the presence of amorphous spherical particles (Fig. 2b). TEM and STEM showed that these spherical particles produced the characteristic two-line diffraction pattern consistent with ferrihydrite (Fig. 2c). The SEM-EDS results revealed that the spherical particles contained approximately 10 at% Si (SI Table 1). These particles were located at the interface between the Fe-oxide powder and the surrounding soil, suggesting that the Si originated from silicate anions released via the chemical weathering of silicate clay minerals in the soil (Schulz et al. 2022).

The XRD, SEM, SEM-EDS, and TEM/STEM results indicated that the Fe-oxide powder contained goethite, lepidocrocite, and ferrihydrite and incorporated Si derived from the surrounding soil. These findings suggest that contact with the surrounding soil matrix affected the Fe-oxide minerals formed from the Fe powder. This highlights the need to investigate the composition of Fe-oxide minerals in the Fe-oxide powder and in the adjacent soil, considering the influence of soil characteristics.

Identifying Fe-oxide minerals formed inside soil adjacent to Fe-oxide powder

A thin-section sample (SI Fig. 1) containing the Fe-oxide powder and the surrounding soil was analyzed to investigate the influence of the contact of the Fe powder with the soil on the formation of Fe-oxide minerals from the Fe powder. Elemental mapping of the thin-section samples revealed that the highest Fe concentrations occurred immediately adjacent to the Fe powder and gradually decreased with increasing distance from the powder (Fig. 3a (A)–(C)). The

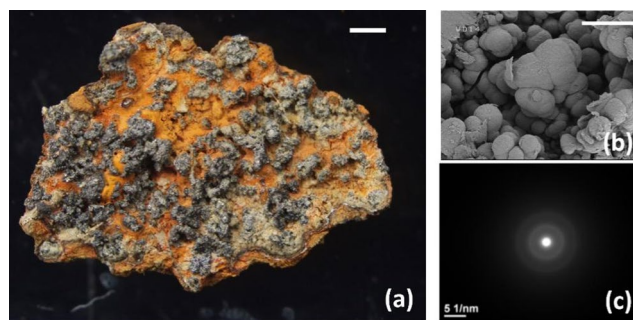


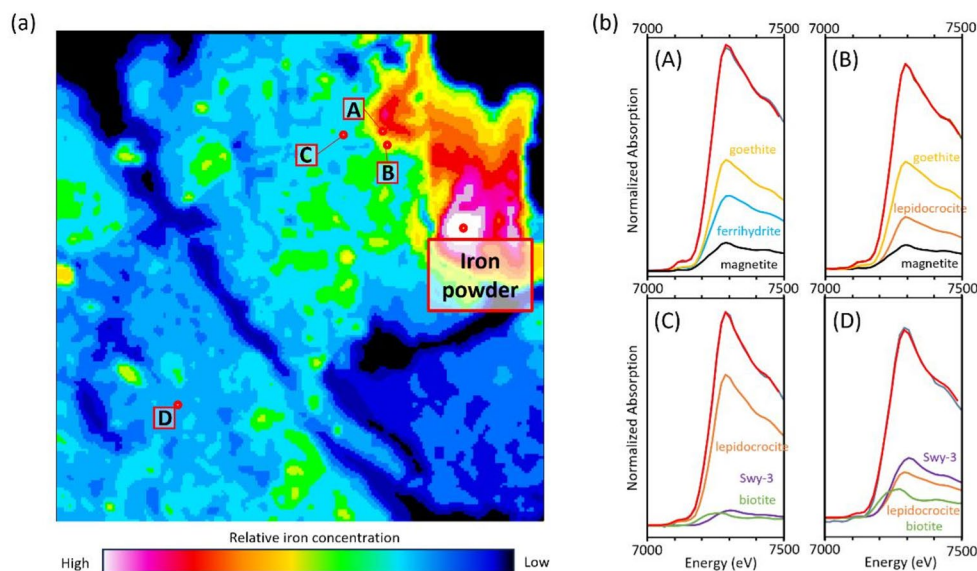
Fig. 2 Fe-oxide powder images obtained using optical microscopy (a); spherical crystal images obtained using SEM (b) and TEM (c). Scale bars indicate 1 mm (a), 3 μ m (b), and 5 1/nm (c)

Fe concentrations declined to levels comparable to those of the natural soil at approximately 2 mm from the Fe powder (Fig. 3a (D)), suggesting that Fe-oxide minerals were generated from the powder formed within 2 mm of the source.

XANES measurements were conducted in areas exhibiting site-specific Fe concentrations, providing estimates of the differences in Fe mineral composition (Fig. 3b). Within 0.1 mm of the Fe powder, the estimated goethite: ferrihydrite: magnetite ratio was approximately 5:3:1. (Fig. 3b (A)), and the goethite: lepidocrocite: biotite ratio was approximately 6:3:1 (Fig. 3b (B)). Lepidocrocite was dominant at approximately 2 mm from the powder (Fig. 3b (C)). Only lepidocrocite and clay minerals were found in areas > 2 mm from the Fe powder (Fig. 3b (D)).

The ferrihydrite detected near the Fe powder may have initially formed via the oxidation of zero-valent or dissolved divalent Fe and rapidly transformed into goethite under high-oxygen conditions (Liu et al. 2022). Ferrihydrite and goethite likely co-occurred near the Fe powder because of the high oxygen concentrations at the soil surface. Magnetite

Fig. 3 Mapping of Fe in the thin sample. **a** The Fe concentration decreased as follows: white > red > yellow > green > blue. “Iron powder” indicates areas where Fe powder was attached. **b** A–D indicated with red circles were analyzed using μ XAFS. Blue and red lines indicate measured and fitted spectra, respectively



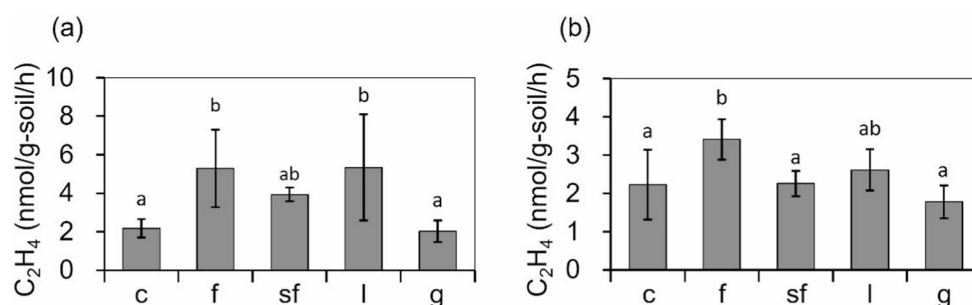


Fig. 4 N-fixing (acetylene reduction) activity of paddy soil microcosms after 6 (**a**) and 9 (**b**) days of incubation under control conditions (no Fe added) or with (c), ferrihydrite (f), Si-containing ferrihydrite (sf), lepidocrocite (l), and goethite (g). Error bars represent standard

deviation ($n=3$), and different letters indicate significant differences between conditions determined using Tukey's test at the 5% probability level

was also observed in this region and was likely present in the original Fe powder prior to oxidation (Fig. 1b).

Lepidocrocite was more abundant at larger distances from the Fe powder, which may have been formed because crystallization was inhibited during the transformation of ferrihydrite to goethite. Ferrihydrite is more likely to transform into lepidocrocite than goethite at high Si concentrations (Schulz et al. 2022). Therefore, the increasing proportion of lepidocrocite with increasing distance from the Fe powder (approximately 0.1–2 mm) was attributed to interactions with Si-rich soils.

The SEM and TEM/STEM results revealed that the ferrihydrite near the Fe powder contained soil-derived Si (Fig. 2b), suggesting that Si-containing ferrihydrite was more likely to form in areas with more soil interactions. However, the μ XRF-XAFS results showed that lepidocrocite was the dominant Fe-oxide mineral formed in areas farther from the Fe powder (Fig. 3 (C)). One explanation for this is that the synthesized Si-containing ferrihydrite used as a standard in this analysis did not fully capture the diversity of Si in the soil or the structural variability of naturally occurring ferrihydrite. Although the ferrihydrite identified using SEM/TEM contained approximately 10% Si (SI Table 1), ferrihydrites with a broader range of Si content and crystallographic structures may exist in the soil.

The SEM, TEM/STEM, and μ XRF-XAFS results indicate that ferrihydrite and goethite were the predominant Fe-oxide minerals in the immediate vicinity of the Fe-oxide powder, whereas lepidocrocite was the dominant Fe-oxide mineral in the soil slightly farther away from the Fe powder, where lepidocrocite formation was strongly influenced by the presence of Si.

Increase in soil N-fixing activity caused by each Fe-oxide mineral

The above analysis suggests that the Fe minerals in the Fe-oxide powder, that is, goethite, lepidocrocite, ferrihydrite,

and Si-containing ferrihydrite, contributed to the enhanced N-fixing activity, likely through the activity of Fe-reducing bacteria in paddy soils, as previously observed (Zhang et al. 2023). Biotite and montmorillonite were not selected as added Fe minerals because they are native soil minerals, rather than Fe oxides derived from elemental Fe powder. Similarly, magnetite was not selected as an added Fe mineral because it was judged to have been originally present in the elemental Fe powder. Paddy soil microcosms were constructed, amended with each of these minerals, and incubated for 9 days, with N-fixing activity measured on days 6 and 9 to estimate the contribution of each Fe mineral to the enhancement of N fixation. The N-fixing activity of the soil amended with ferrihydrite (5.3 ± 2.0 and 3.4 ± 0.5 nmol/g soil/h on days 6 and 9, respectively) was substantially higher than that of the control soil (3.9 ± 0.4 and 2.3 ± 0.3 nmol/g soil/h on days 6 and 9, respectively) (Fig. 4). The N-fixing activity of the soil amended with lepidocrocite was markedly higher (5.3 ± 2.8 nmol/g soil/h) than that of the control soil on day 6 (Fig. 4a), although the difference was not significant on day 9 (Fig. 4b). In contrast, the N-fixing activity of soils amended with goethite or Si-containing ferrihydrite did not differ considerably from that of the control at either time point. These results indicate that the addition of ferrihydrite and lepidocrocite enhanced the soil N-fixing activity. Based on our previous study, this activity was suggested to be primarily derived from the Fe-reducing bacteria.

Generally, the larger the specific surface area of an Fe mineral, the greater its potential accessibility to microorganisms (Roden and Zachara 1996). Similarly, in the present study, the highly crystalline mineral goethite did not enhance soil N-fixing activity to the same extent as ferrihydrite and lepidocrocite, which have low crystallinity. In previous studies, the addition of ferrihydrite to paddy soil microcosms increased the abundance of the nitrogenase gene (*nifD*) transcripts in Fe-reducing bacteria (Masuda et al. 2024a). Similarly, the addition of low-crystallinity Fe minerals likely enhanced the N-fixing activity of Fe-reducing bacteria.

Conversely, despite being a low-crystallinity Fe mineral with large specific surface areas, ferrihydrite showed a more sustained effect on N fixation, whereas that of lepidocrocite was slightly diminished by day 9. These two Fe oxides differ in specific surface area by approximately one order of magnitude (Larsen and Postma 2001), which may have contributed to the observed differences in biological activity. In addition, under reducing conditions, lepidocrocite may have transformed into magnetite, which has a lower bioavailability to microorganisms (Liu et al. 2022). Furthermore, Si-containing ferrihydrite, owing to its low crystallinity (Childs 1992), should enhance the N-fixing activity similar to that of ferrihydrite. However, in this study, Si-containing ferrihydrite did not enhance the N-fixing activity. This may be due to the formation of a Si coating on the ferrihydrite surface (Dyer et al. 2010). Our XRD results showed that the diffraction peaks of ferrihydrite broadened with increasing Si content (SI Fig. 2), indicating the structural incorporation of Si^{4+} (Boyd and Scott 1999; Cismasu et al. 2014), which may have hindered microbial access and decreased the bioavailability of ferrihydrite.

These results indicate that low crystallinity alone is insufficient for Fe-oxide minerals to enhance biological N fixation in paddy soils. Future studies must test these hypotheses by elucidating more direct evidence of crystallinity, crystal structure, and microbial accessibility to better understand the relationship between Fe minerals and microbial activity.

Conclusion

We characterized the crystal structures of Fe oxides generated from pure (zero-valent) Fe powder applied to paddy soils. Next, we evaluated the influence of different Fe-oxide morphologies on soil N-fixing activity, which we previously showed to be primarily associated with Fe-reducing bacteria. The results showed that goethite, lepidocrocite, ferrihydrite, and Si-containing ferrihydrite were generated from Fe powder in paddy soils, and among these minerals, only lepidocrocite and ferrihydrite enhanced the N-fixing activity. These findings suggest that ferrihydrite and lepidocrocite are the Fe-oxide minerals most likely responsible for the enhanced N fixation associated with the Fe-reducing bacteria observed in our previous field study. Fe oxidation should be optimized to selectively produce Fe oxides and enhance N fixation in paddy soils through Fe application.

The results of this study will enable more precise control of microbial N fixation in soils and indicate its feasibility for sustaining rice yields. However, whether the applied Fe powder undergoes similar transformations in soils under environmental conditions or in soil types different from those in this study remains unclear. The formation of Si-containing

ferrihydrite suggests that in soils rich in silicate minerals, Fe application may not sufficiently enhance N-fixing activity. Future studies should identify the environmental factors controlling Fe mineral transformations to establish Fe application as a novel strategy for enhancing N fixation, reducing the dependence on synthetic N fertilizers, and mitigating the environmental burden of excessive N input.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s10333-026-01103-5>.

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